

PROJECT: Custom VPC setup with Bastion Host & NAT Gateway

Services: VPC, EC2, NAT Gateway, Bastion Host, Security Groups.

Project Goal: Build a secure private/public subnet architecture with internet access for private instances.

Skills: Routing Tables, Subnetting, SSH Tunnelling.

Definition: A custom VPC setup with a Bastion Host and NAT Gateway is an AWS network architecture where a user defined Virtual Private Cloud is created with public and private subnets, a Bastion Host is deployed in the public subnet to provide secure administrative access to private instances, and a NAT Gateway is used to allow private instances outbound internet access without exposing them to inbound internet traffic.

Scope of the Project: The scope of this project is to design and implement a secure and scalable AWS network architecture using a custom Virtual Private Cloud (VPC) that separates resources into public and private subnets.

The project focuses on providing:

- Secure administrative access to provide EC2 instance using a Bastion Host.
- Outbound internet connectivity for private instances using a NAT Gateway.
- Strong network isolation and access control using security groups and route tables.

Methodology:

1. Create VPC
2. Create 2 subnets:
 - Public
 - Private
3. Attach Internet Gateway to public subnet
4. Create NAT Gateway for private subnet
5. Launch:
 - Bastion in public
 - Private EC2 in private

Services Used in this Project:

1.VPC (Virtual Private Cloud)

Service: Amazon VPC

- Creates your own private network
- You define IP range (CIDR)

2. Subnets

Part of VPC

- Public Subnet → internet access
- Private Subnet → no direct internet

3. EC2 (Virtual Servers)

Service: **Amazon EC2**

- Bastion Host (in public subnet)
- Private Server (in private subnet)

4. Internet Gateway

Service: **Internet Gateway**

- Allows public subnet to connect to internet

5. NAT Gateway

Service: **NAT Gateway**

- Allows private servers to access internet
- Blocks incoming internet traffic

6. Route Tables

Service: **VPC Route Tables**

- Controls traffic flow

Example:

- Public → IGW
- Private → NAT

7. Security Groups

Service: **Security Groups**

- Controls inbound & outbound traffic

8. Key Pair

Service: **EC2 Key Pair**

- Used to login (SSH) to instances

9. Elastic IP

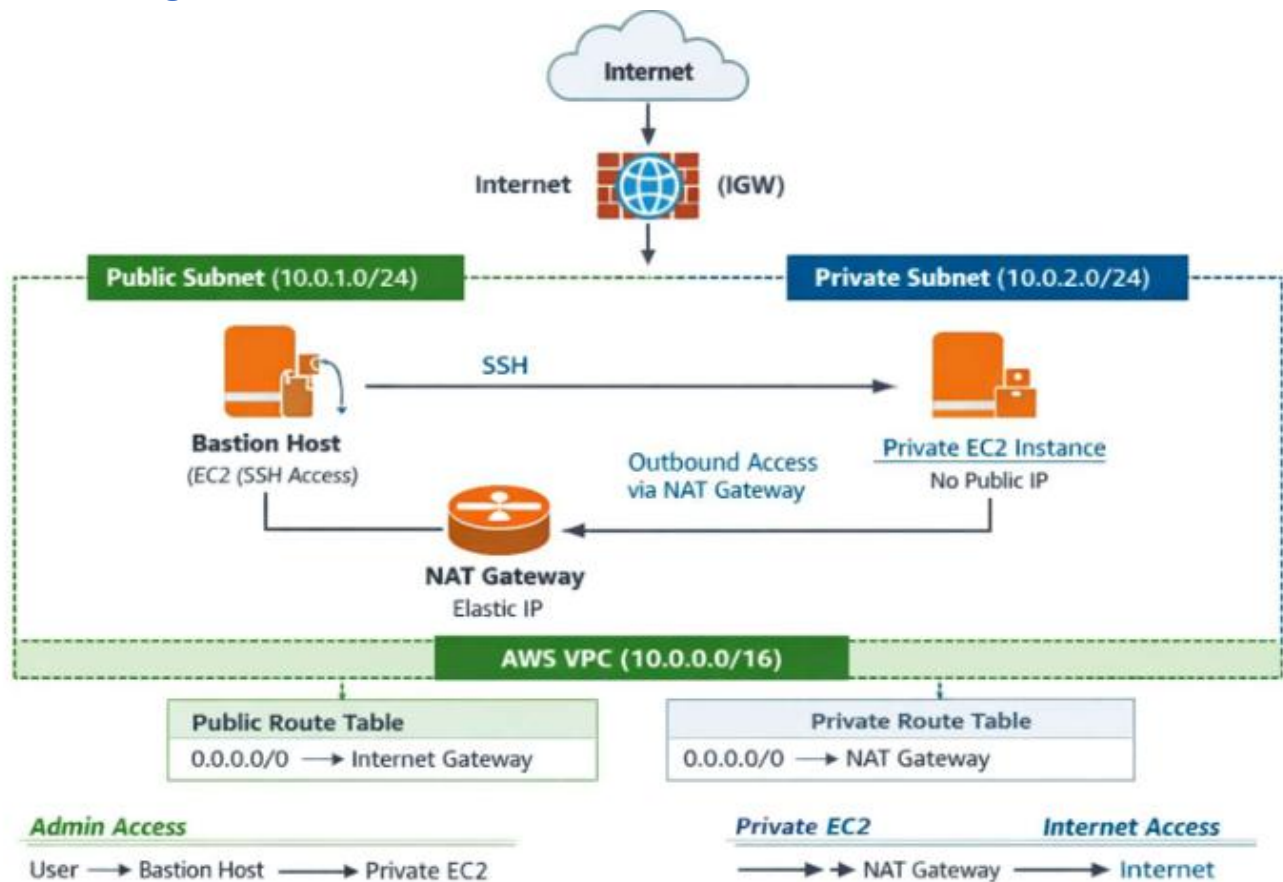
Service: **Elastic IP**

- Used for:
 - Bastion Host
 - NAT Gateway

10. SSH (Secure Shell)

Entity: Secure Shell (SSH)

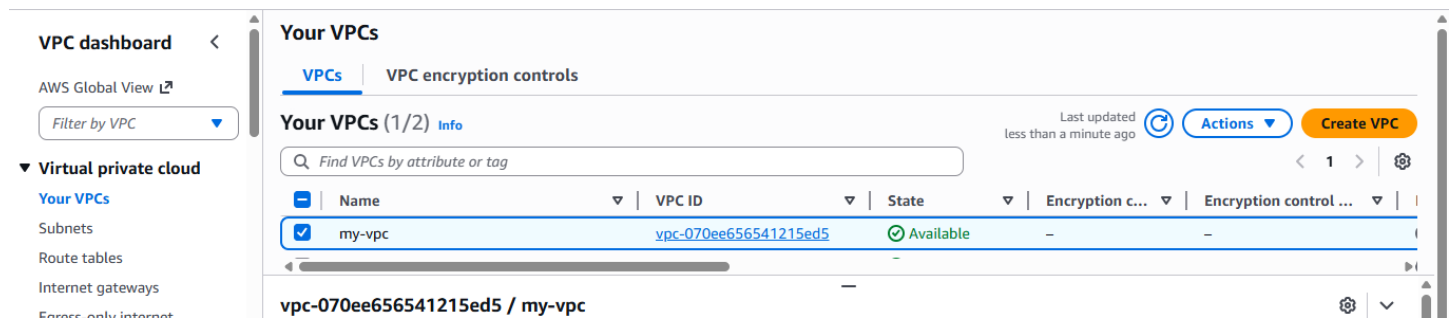
Architecture Diagram:



Implementation and Execution of a custom VPC setup with Bastion Host and NAT Gateway.

Phase 1: Create VPC.

- Open AWS Management Console
- Click create VPC
- Select VPC only
- Configure: -Name: my-vpc-IPV4 CIDR: 10.0.0.0/16 -Tenancy: Default
- Click create VPC



Phase 2: Subnet:

- Go to Subnets --Create Subnet
- Select VPC: my-vpc
- Subnet Name:Public-Subnet1
- Availability Zone: ap-south-1a
- IPV4 CIDR: 10.0.1.0/24
- Click create subnet.

The screenshot shows the AWS VPC console 'Subnets' page. On the left, the 'VPC dashboard' sidebar includes 'Virtual private cloud', 'Your VPCs', 'Subnets' (selected), and 'Route tables'. The main content area is titled 'Subnets (1/1) Info' and shows a search bar with 'subnet-0681e58d9e62d1c91' and a 'Clear filters' button. Below the search bar is a table with columns: Name, Subnet ID, State, VPC, and Block Public. The table contains one entry: 'Public-Subnet-1' with Subnet ID 'subnet-0681e58d9e62d1c91', State 'Available', VPC 'vpc-070ee656541215ed5 | my-...', and Block Public 'Off'. At the top right, there are buttons for 'Actions' and 'Create subnet', and a note 'Last updated 4 minutes ago'.

Private Subnet:

- Go to Subnets--Create Subnet
- Subnet Name: Private-Subnet1
- Availability Zone: ap-south-1a
- IPV4 CIDR: 10.0.2.0/24
- Click create Subnet

The screenshot shows the AWS VPC console 'Subnets' page. On the left, the 'VPC dashboard' sidebar includes 'Virtual private cloud', 'Your VPCs', 'Subnets' (selected), and 'Route tables'. The main content area is titled 'Subnets (1/1) Info' and shows a search bar with 'subnet-0b1b15839c24d2b36' and a 'Clear filters' button. Below the search bar is a table with columns: Name, Subnet ID, State, VPC, and Block Public. The table contains one entry: 'Private-Subnet-1' with Subnet ID 'subnet-0b1b15839c24d2b36', State 'Available', VPC 'vpc-070ee656541215ed5 | my-...', and Block Public 'Off'. At the top right, there are buttons for 'Actions' and 'Create subnet', and a note 'Last updated 5 minutes ago'.

Phase 3: Create Internet Gateway (IGW)

- Go to Internet Gateway ? Create
- Name: my-IGW
- Click create Internet Gateway
- Select IGW ? Actions ? Attach to VPC
- Choose Bastion-VPC
- Click on Attach.

VPC > Internet gateways

Internet gateways (1/1) Info

Find internet gateways by attribute or tag

☒	Name	Internet gateway ID	State	VPC ID
☒	My-IGW	igw-0bb3ce0812aed504d	Attached	vpc-070ee656541215ed5 my-vpc

Actions Create internet gateway

Phase 4: Create Public Route Table Public Route Table:

- Go to Route Tables---Create route table
- Name: Public-RT
- VPC: my-vpc
- Click create route table Add Route:
- Select Bastion-Public-RT1
- Routes: Edit routes
- Add routes: — Destination: 0.0.0.0/0 — Target: Internet Gateway (Bastion-IGW)
- Save routes

VPC dashboard <

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Route tables (1/5) Info

Last updated 8 minutes ago

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-082a075cf23868ef0	-	nat-18a22c4ed954...	No	vpc
☐	-	rtb-02a97314e232a2ed8	-	-	Yes	vpc
☐	-	rtb-019cec7cbd5215f3f	-	-	Yes	vpc
☒	Public-RT	rtb-0544a12af615dabc0	2 subnets	-	No	vpc

rtb-0544a12af615dabc0 / Public-RT

Details Routes Subnet associations Edge associations Route propagation Tags

Associate Public Subnet:

- Subnet Associations: Edit
- Select Public-Subnet1
- Save

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

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Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Route tables (1/5)

Find route tables by attribute or tag

Name

Route table ID

Explicit subnet associ...

Edge associations

Main

VPC

Public-RT

rtb-0544a12af615dabc0

2 subnets

-

No

vpc

rtb-0544a12af615dabc0 / Public-RT

Find subnet association

Name

Subnet ID

IPv4 CIDR

IPv6 CIDR

Public-Subnet-1

subnet-0681e58d9e62d1c91

10.0.1.0/24

-

Phase 5: Create NAT Gateway

- Go to NAT Gateway---Create
- Name: my-NAT-GATEWAY
- Subnet: Public-Subnet1
- Connectivity type: Public
- Elastic IP: Allocate Elastic IP
- Click create NAT Gateway
- Wait until status becomes Available

Filter by VPC

Virtual private cloud

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Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

NAT gateways (1/1)

Find NAT gateways by attribute or tag

Name

NAT gateway ID

Connectivity...

State

State message

Availability ...

my-NAT-GATEWAY

nat-18a22c4ed954e8d51

Public

Available

-

Regional

nat-18a22c4ed954e8d51 / my-NAT-GATEWAY

Associated IP addresses (1)

IP address

Status

Availability Zone

Allocation ID

3.109.118.27

Succeeded

aps1-az1 (ap-south-1a)

eipalloc-01ffc99e

Filter by VPC

vpc-070ee656541215ed5

my-vpc

Owner: 074139044318

Virtual private cloud

Your VPCs

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Route tables

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Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Elastic IP addresses (1/1)

Find elastic IP addresses by attribute or tag

15.206.112.158

Clear filters

Allocated IPv4 addr...

Type

Allocation ID

Reverse DNS record

Associated instance ID

15.206.112.158

Public IP

eipalloc-060b0691d507b317e

-

-

15.206.112.158

View IP address usage and recommendations to release unused IPs with [Public IP Insights](#)

Summary

Tags

Phase 6: Create Private Route Table:

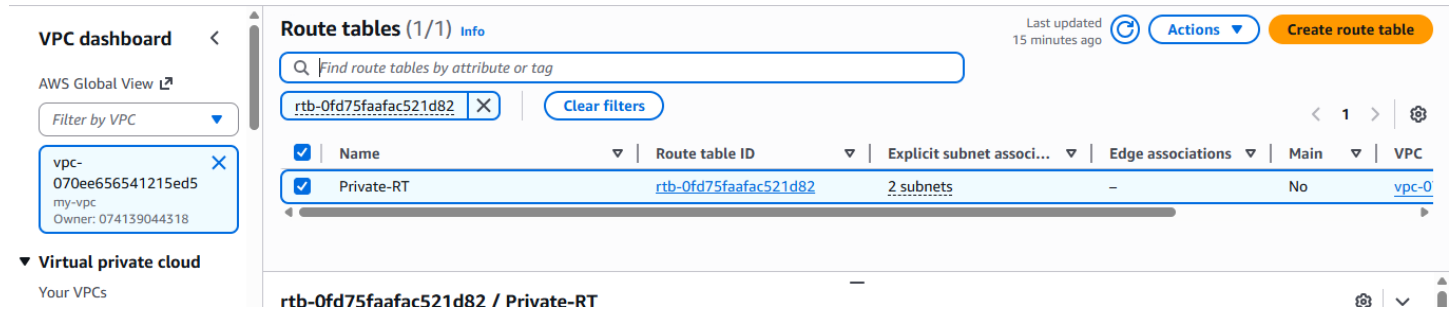
Private Route Table:

Go to Route Tables---Create

Name: Bastion-Private-RT1

VPC: Bastion-VPC

Click create



VPC dashboard < AWS Global View [L](#)

Filter by VPC [v](#)

vpc-070ee656541215ed5 my-vpc Owner: 074139044318

▼ **Virtual private cloud**
Your VPCs

Route tables (1/1) Info Last updated 15 minutes ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

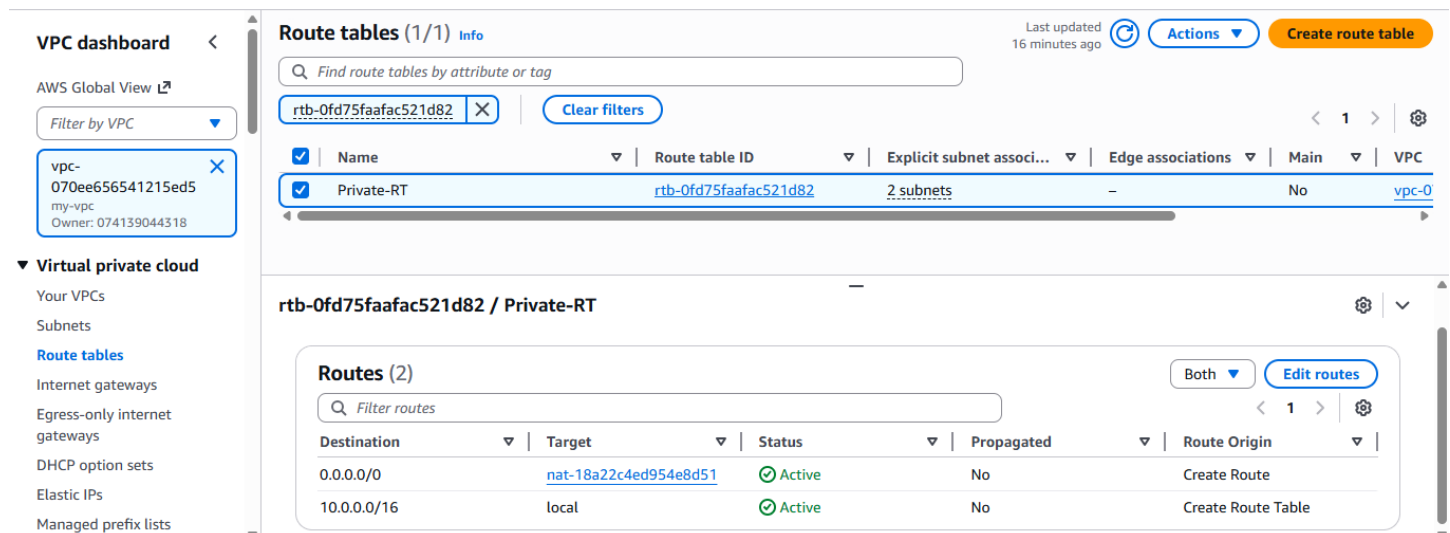
rtb-0fd75faafac521d82 [Clear filters](#)

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	Private-RT	rtb-0fd75faafac521d82	2 subnets	–	No	vpc-0

rtb-0fd75faafac521d82 / Private-RT

Add NAT Route:

- Select Bastion-Private-RT1
- Routes: Edit routes
- Add: → Destination: 0.0.0.0/0 → Target: NAT Gateway (Private NAT Gateway1)
- Save



VPC dashboard < AWS Global View [L](#)

Filter by VPC [v](#)

vpc-070ee656541215ed5 my-vpc Owner: 074139044318

▼ **Virtual private cloud**
Your VPCs
Subnets
[Route tables](#)
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists

Route tables (1/1) Info Last updated 16 minutes ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

rtb-0fd75faafac521d82 [Clear filters](#)

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	Private-RT	rtb-0fd75faafac521d82	2 subnets	–	No	vpc-0

rtb-0fd75faafac521d82 / Private-RT

Routes (2) [Both](#) [Edit routes](#)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-18a22c4ed954e8d51	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Associate Private Subnet:

- Subnet associations: Edit
- Select: Private-Subnet-1
- Save

VPC dashboard < AWS Global View [↗](#)

Filter by VPC [▼](#)

vpc-070ee656541215ed5
my-vpc
Owner: 074139044318

▼ **Virtual private cloud**

Your VPCs
Subnets
Route tables
Internet gateways

Route tables (1/4) Info

Find route tables by attribute or tag

VPC: vpc-070ee656541215ed5 [✕](#) [Clear filters](#)

Last updated 18 minutes ago [🔄](#) [Actions](#) [Create route table](#)

	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	Public-RT	rtb-0544a12af615dabc0	2 subnets	-	No	vpc
☒	Private-RT	rtb-0fd75faafac521d82	2 subnets	-	No	vpc

rtb-0fd75faafac521d82 / Private-RT

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Private-Subnet-1	subnet-0b1b15839c24d2b36	10.0.3.0/24	-

Phase 7: Create Security Group

Bastion Host Security Group:

- Go to Security Group [🔗](#) Create
- Name: Bastion-securitygroup
- VPC: Bastion-VPC
- Inbound Rule:
- Type: SSH
- Port Range: 22
- Source: Anywhere IPv4 (0.0.0.0)

Security Groups (1/4) Info

Find security groups by attribute or tag

sg-08e0ca9f94873ff37 [✕](#) [Clear filters](#)

[🔄](#) [Actions](#) [Export security groups to CSV](#) [Create security group](#)

	Name	Security group ID	Security group name	VPC ID
☐	-	sg-047a2b1b00763ef82	private-securitygroup	vpc-070ee656541215ed5
☒	-	sg-08e0ca9f94873ff37	Bastion-Securitygroup	vpc-070ee656541215ed5
☐	-	sa-01364663254dfbfd5	launch-wizard-11	vpc-070ee656541215ed5

sg-08e0ca9f94873ff37 - Bastion-Securitygroup

Inbound rules (1/2)

Search

	Name	Security group rule ID	IP version	Type	Protocol
☐	-	sgr-0674e3b85269d3d66	IPv4	HTTP	TCP
☒	-	sgr-00ede136dbc7e37b0	IPv4	SSH	TCP

- Outbound Rule:
- Type: All traffic
- Port Range: All

- Destination: 0.0.0.0/0
- Click create

Security Groups (1/4) Info

Find security groups by attribute or tag

sg-08e0ca9f94873ff37 Clear filters

	Name	Security group ID	Security group name	VPC ID
☐	-	sg-047a2b1b00763ef82	private-securitygroup	vpc-070ee656541215ed5
☒	-	sg-08e0ca9f94873ff37	Bastion-Securitygroup	vpc-070ee656541215ed5
☐	-	sa-01364663254dfbfd5	launch-wizard-11	vpc-070ee656541215ed5

sg-08e0ca9f94873ff37 - Bastion-Securitygroup

Outbound rules (1/1)

Search

	Name	Security group rule ID	IP version	Type	Protocol
☒	-	sgr-0945c63ad2c9437c5	IPv4	All traffic	All

Phase 8: Create Private EC2 Security Group

Private EC2 Security Group:

- Create Security Group
- Name: Private-Securitygroup
- VPC: my-vpc
- Inbound Rule: —
- Type: SSH, All ICMP
- Source: Bastion-Securitygroup (Bastion-Host)

Security Groups (1/4) Info

Find security groups by attribute or tag

sg-08e0ca9f94873ff37 Clear filters

	Name	Security group ID	Security group name	VPC ID
☒	-	sg-047a2b1b00763ef82	private-securitygroup	vpc-070ee656541215ed5
☐	-	sg-08e0ca9f94873ff37	Bastion-Securitygroup	vpc-070ee656541215ed5
☐	-	sa-01364663254dfbfd5	launch-wizard-11	vpc-070ee656541215ed5

sg-047a2b1b00763ef82 - private-securitygroup

Inbound rules (1/1)

Search

	Name	Security group rule ID	IP version	Type	Protocol
☒	-	sgr-0e4e5ae06b95ec80f	-	SSH	TCP

- Outbound Rule:
 - Type: All traffic
 - Destination: 0.0.0.0/0
 - Click create

VPC > Security Groups

Route tables
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Egress-only internet gateways
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Elastic IPs
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NAT gateways
Peering connections
Route servers

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Network ACLs
Security groups

▼ PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services

Security Groups (1/4) Info

Find security groups by attribute or tag

sg-08e0ca9f94873ff37 [Clear filters](#)

	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-055cf3c03d122d88e	launch-wizard-5	vpc-070ee656541215ed5	launch-wizard-5
☒	-	sg-047a2b1b00763ef82	private-securitygroup	vpc-070ee656541215ed5	Private EC2 acce
☐	-	sg-08e0ca9f94873ff37	Bastion-Securitygroup	vpc-070ee656541215ed5	Allow SSH acces
☐	-	sg-01364663254dfbfd5	launch-wizard-11	vpc-070ee656541215ed5	launch-wizard-1

sg-047a2b1b00763ef82 - private-securitygroup

Details | **Inbound rules** | **Outbound rules** | Sharing | VPC associations | Related resources - new | Tags

Outbound rules (1/1)

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range
☒	-	sgr-0c3a78085d28d450b	IPv4	All traffic	All	All

Phase 9: Launch Bastion Host (Public EC2 Instance)

- Go to EC2---Launch instance
- Name: Bastion-Server
- AMI: Amazon Linux Kernel 6.1 2023
- Instance Type: t3. micro
- Key-Pair: Select/Create (Bastion-Key)
- Network Settings:
- VPC: my-vpc
- Subnet: Public-Subnet-1-ap-south-1a
- Auto-assign Public IP: Enable
- Security Group: Create Bastion-Securitygroup
- Launch instance

EC2

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Instance Types
Launch Templates

Instances (1/1) Info

Last updated less than a minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set [Find Instance by attribute or tag \(case-sensitive\)](#)

Instance ID = i-01cebe546e4c058e2 [Clear filters](#)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☒	bastion-server	i-01cebe546e4c058e2	Running	t3.micro	3/3 checks passed	View alarms	ap-south-1a	-

EC2

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Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
AMIs
AMI Catalog

Instance summary for i-01cebe546e4c058e2 (bastion-server) Info

Updated less than a minute ago

Instance ID

i-01cebe546e4c058e2

IPv6 address

-

Hostname type

IP name: ip-10-0-1-53.ap-south-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

65.0.101.181 [Public IP]

IAM role

-

IMDSv2

Required

Public IPv4 address

65.0.101.181 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-1-53.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-070ee656541215ed5 (my-vpc)

Subnet ID

subnet-0681e58d9e62d1c91 (Public-Subnet-1)

Instance ARN

arn:aws:ec2:ap-south-1:074139044318:instance/i-01cebe546e4c058e2

Private IPv4 addresses

10.0.1.53

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Connect

Instance state

Actions

Phase 10: Launch Bastion Host (Public EC2 Instance)

- Go to EC2---Launch instance
- Name: Private-Server
- AMI: Amazon Linux Kernel 6.1 2023
- Instance Type: t3. micro
- Key-Pair: Select/Create (Bastion-Key)
- Network Settings:
- VPC: my-vpc
- Subnet: Private-Subnet-1-ap-south-1a
- Auto-assign Public IP: Disable
- Security Group: Create Private-Securitygroup
- Launch instance

EC2

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Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance ID = i-0deedb8c55da72b7b

Clear filters

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public
☒	private-server	i-0deedb8c55da72b7b	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	-

i-0deedb8c55da72b7b (private-server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary Info

Instance ID

i-0deedb8c55da72b7b

IPv6 address

-

Public IPv4 address

-

Instance state

Running

Private IPv4 addresses

10.0.3.231

Public DNS

-


```
aws [Search] [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Shreenivasa (0741-3904-4318) Shreenivasa
ec2-user@10.0.1.53: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-10-0-1-53 ~]$ sudo su
[root@ip-10-0-1-53 ec2-user]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-10-0-1-53 ec2-user]# curl ifconfig.me
Moved Permanently. Redirecting to https://ifconfig.me/[root@ip-10-0-1-53 ec2-user]#
[root@ip-10-0-1-53 ec2-user]# curl ifconfig.me
65.0.101.181[root@ip-10-0-1-53 ec2-user]# ping google.com
PING google.com (142.250.206.110) 56(84) bytes of data.
64 bytes from dell1s20-in-f14.1e100.net (142.250.206.110): icmp_seq=1 ttl=117 time=2.07 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=2 ttl=117 time=2.10 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=3 ttl=117 time=2.10 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=4 ttl=117 time=2.09 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=5 ttl=117 time=2.11 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=6 ttl=117 time=2.11 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=7 ttl=117 time=2.10 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=8 ttl=117 time=2.11 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=9 ttl=117 time=2.10 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=10 ttl=117 time=2.18 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=11 ttl=117 time=2.12 ms
64 bytes from lcboma-az-in-f14.1e100.net (142.250.206.110): icmp_seq=12 ttl=117 time=2.11 ms
```

Troubleshooting:

In this custom VPC project for me ssh was created in wrong way instead of giving source type Anywhere IPv4 I gave myIP its not working .

Solution: later I changed to Anywhere IPv4 in the source type. It worked properly.

Conclusion:

The custom VPC setup with Bastion Host and NAT Gateway successfully demonstrates the implementation of a secure and well-structured AWS network architecture. By segregating resources into public and private subnets, the project ensures strong network isolation and enhanced security. The Bastion Host acts as a controlled entry point for administrative access, while the NAT Gateway enables private EC2 instances to access the internet securely without being exposed to inbound traffic. Proper configuration of route tables and security groups ensures efficient traffic flow and adherence to AWS best practices. Overall, this project provides hands-on experience in designing, deploying, and managing a scalable and secure cloud infrastructure suitable for real-world production environments.